

ARTICLE · ESG ASSURANCE BRIEFING

AI in ESG Disclosure and Assurance

A guide to audit-ready reporting: how AI is used across the disclosure lifecycle, what CSRD, ISSB, and California now require, and what "audit-ready" actually means for an AI pipeline.

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The full briefing — the 90-day readiness roadmap, the six-control checklist, and the CSRD/ISSB/SB 253 coverage — as one file to share with your team or auditor.

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Sustainability reporting has moved, in the space of three years, from a voluntary communications exercise into a regulated discipline that increasingly sits alongside financial reporting. Three regimes anchor that shift: the European Union's Corporate Sustainability Reporting Directive (CSRD), the International Sustainability Standards Board's IFRS S1 and S2, and, in the United States, California's SB 253 and SB 261. Every one of these regimes has been reshaped by rulemaking through 2026, and every one of them now expects sustainability data to be traceable, consistent, and ready for third-party assurance in the same way financial statements are. Artificial intelligence has become central to meeting that expectation — not chiefly because it drafts faster, but because, used correctly, it can make sustainability data reconstructible from source document to final disclosure. This briefing sets out how AI is being applied across the disclosure lifecycle, what recent evidence says about its use by investors and researchers, what "audit-ready" actually requires of an AI pipeline, and how companies connected to EU and multinational supply chains — including PSX-listed and Pakistani exporting companies — should prepare over the next two quarters.

WHERE THE RULES STAND IN 2026

The regulatory picture has narrowed, not disappeared

The CSRD's simplification package, known as Omnibus I, is no longer a proposal. Directive (EU) 2026/470 was published in the Official Journal on 26 February 2026 and entered into force on 18 March 2026. It cuts the mandatory scope of the CSRD by roughly four-fifths: reporting is now required only from EU undertakings with more than 1,000 employees and more than €450 million in net turnover, and from non-EU groups earning more than €450 million in EU turnover with a qualifying EU subsidiary or branch. Listed SMEs are exempt outright, sector-specific ESRS standards have been dropped, and Wave 2 companies now report for financial year 2027. Limited assurance — not the previously planned move to reasonable assurance — remains the standard, and double materiality remains the methodological foundation of the whole exercise. In practical terms, the population of companies with a direct, mandatory CSRD reporting obligation has fallen from an estimated 50,000 across the EU to roughly 5,000.

50,000	~5,000	€450m
Companies in mandatory CSRD scope before Omnibus I	Companies in scope after Omnibus I (effective 18 Mar 2026)	Net turnover threshold for a mandatory obligation

That narrower scope has not reduced the pressure on data quality for the companies that remain in scope, and it has created a new obligation that matters for suppliers in emerging markets: the value chain cap. Omnibus I extends this protection to any company with 1,000 employees or fewer, EU or non-EU. A CSRD-reporting customer cannot require such a supplier to provide sustainability information beyond what the forthcoming Voluntary SME Standard (VSME) covers, and the supplier has a statutory right to decline more extensive requests. The European Commission is required to publish the VSME content by July 2026.

Outside the EU, the ISSB's IFRS S1 and S2 continue to gain jurisdictional traction even as the CSRD contracts. As of 22 April 2026, S&P Global counted 28 jurisdictions that had adopted the ISSB standards on a voluntary or mandatory basis, up from 21 at the start of the year, with a further 12 planning adoption. Markets that have moved include Japan, Australia, Singapore, Hong Kong, South Korea, Canada, the UK, Brazil, Nigeria, Turkey, Chile, Qatar, and Mexico; China is developing an aligned but locally adapted standard targeting a 2030 rollout. The United States has not adopted ISSB standards federally, but California's own climate laws continue to move forward independently. SB 253 requires companies with more than \$1 billion in revenue doing business in California to report Scope 1 and Scope 2 emissions; the deadline has been pushed back twice in 2026, most recently to 10 November 2026. SB 261, covering companies above \$500 million in revenue, has a statutory deadline of 1 January 2026 for biennial climate-risk reports, though CARB has paused enforcement pending a Ninth Circuit appeal.

The overall picture is not deregulation — it is triage. Fewer companies face a direct EU statutory obligation, but the ones that remain in scope, together with the growing list of ISSB-adopting jurisdictions and the California regime, face reporting expectations that assume auditable, version-controlled data. That is the environment AI is being deployed into.

WHY THIS MATTERS COMMERCIALLY

Investors are already reading disclosures with AI

Clarity AI's Investor Sustainability Pulse 2025, a survey of more than 120 asset managers, asset owners, and financial institutions across North America, EMEA, and APAC, found that close to 60% of global investors are already using, or plan to use, AI in sustainability and investment analysis, with trust in model outputs now the main constraint on scaling that use further. The same survey found that only 28.9% of respondent organisations considered themselves fully prepared for 2026 sustainability disclosure requirements, while 56.2% reported being only partially prepared or not prepared at all, and 89% expected cross-border regulatory divergence to complicate their investment strategies. Separately, 55% of respondents said extreme weather events were already influencing investment decisions in real time.

~60% Investors using or planning to use AI in sustainability analysis	28.9% Consider themselves fully prepared for 2026 disclosure rules	89% Expect regulatory divergence to complicate cross-border investing
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The practical consequence is that a company's disclosures are now more likely than not to be read, compared, and cross-checked by a model before a human analyst ever opens the PDF. A narrative that reads well but cannot be reconciled against the underlying data is no longer a stylistic weakness; it is something an automated reader will flag as an inconsistency.

THE DISCLOSURE PIPELINE

Where AI is actually being used across the reporting lifecycle

It helps to think of ESG reporting as a pipeline with five stages. AI can add real value at each one, but each stage also introduces a specific assurance risk if the underlying controls are weak — the two move together.

1

Data ingestion and normalisation

AI tools ingest raw activity data from PDFs, emails, APIs, and ERP exports, extracting meter readings, fuel volumes, and travel segments, and normalising units and timestamps into a consistent structure.

ASSURANCE RISK

If the pipeline cannot point back to the original invoice or feed and show the extraction rule applied, the resulting metric is not verifiable — however precise it looks.

2

Validation, cleaning, and gap-filling

Models detect anomalies such as an unexplained tenfold spike in electricity use, impute missing months using pattern-aware estimation, and enforce basic business rules like disallowing negative consumption figures.

ASSURANCE RISK

Any high-variance imputation needs a documented human sign-off, or an auditor testing that data point is very likely to raise a finding.

3

Calculation, factor selection, and framework mapping

AI-assisted systems select emission factors with a recorded version and source, apply GHG accounting formulas, and map the same underlying metrics to ESRS, ISSB, CDP, or GRI indicators at once.

ASSURANCE RISK

Every reported figure must be reconstructable as a chain: source data, transformation, factor, formula, final value. A break anywhere is a control gap, not a rounding issue.

4

Narrative drafting and disclosure assembly

Generative tools draft indicator narratives and management commentary from live data, in principle preserving a link back to the source metrics that support each sentence.

ASSURANCE RISK

Drafts need source attribution and a versioned record of the prompts used. Without that, an auditor cannot test whether the narrative reflects the data behind it.

5

Assurance preparation and continuous control testing

Mature pipelines run continuous checks on accuracy, drift, and completeness across the year, keep immutable logs, and can export a line-item audit trail on demand.

ASSURANCE RISK

Under CSRD and ISSB-aligned expectations, this evidence needs to already exist before the auditor asks for it. Building it retroactively is exactly what produces limited-assurance qualifications.

Because CSRD's third-party assurance requirement remains in force, AI used anywhere in the sustainability reporting chain needs to be governed with the same discipline as a financial system. In practice, that comes down to six controls.

Immutable data lineage Record the source system, extraction method, transformation steps, and final mapping for every reportable field, in a log that cannot be edited after the fact.	Model and prompt versioning Version every prompt, system instruction, retrieval corpus, and model release used in the pipeline, tied to a named approver and timestamp.	Human-in-the-loop gates Require manual sign-off for any AI-imputed value that exceeds a defined variance threshold — commonly around 5% — before it enters the reported figures.
Continuous validation Schedule accuracy, drift, and data-quality tests throughout the year rather than at year-end, so control evidence already exists.	Service account governance Give pipeline identities minimum access, separate read from write functions, and log every automated action affecting a reported metric.	Exportable audit trail Produce, on request, a step-by-step trail from the raw source document through to the final reported metric, including factor version and formula.

DETECTION, HONESTLY ASSESSED

Can AI actually detect greenwashing?

AI's role here runs in two directions at once. Used well, it helps investors and regulators spot the gap between what a report claims and what the underlying data shows; used carelessly, generative AI can also produce more polished, framework-aligned narrative even when the substance behind it is thin — what researchers call "sophisticated symbolic compliance." Most detection approaches look for specific signals rather than "greenwashing" as a single labelled category:

Text–performance gap: confident, positive language paired with weak or absent measured outcomes.
Unquantified claims: phrases like "significant reduction" with no attached figure.
Scope 3 omission: net-zero pledges that exclude value-chain emissions.
Target–trajectory mismatch: a 2050 net-zero commitment with no credible interim milestones.
Year-over-year inconsistency: contradictions between filing periods.

The technical results are genuinely promising: a 2026 study combining transformer-based NLP with financial-ESG divergence modelling, built for the UK regulatory context, reported 86.34% accuracy identifying greenwashing risk patterns. But a broader 2025–2026 survey covering 61 research works on NLP-based greenwashing

detection found that no publicly available dataset of confirmed positive and negative greenwashing examples currently exists — most systems are trained against proxy indicators rather than verified ground truth. That is a reason to treat any single AI-generated "greenwashing score" as one input among several, not a verdict.

IMPLEMENTATION

A practical architecture for an assurance-ready AI stack

A stack that can survive third-party assurance separates concerns clearly and logs everything that happens between them, rather than treating the whole pipeline as one black box.

INGESTION LAYER	Connectors to ERPs, utility APIs, travel systems, and supplier portals, with OCR/NLP for PDFs and domain-specific validators applied at the point of entry.
DATA QUALITY ENGINE	Anomaly detection, completeness checks, imputation with confidence intervals, and a justification log for every factor or model choice.
CALCULATION & MAPPING	GHG calculations against versioned emission factors, with automatic mapping to ESRS, ISSB, and CDP indicators and audit trail generated as a by-product.
NARRATIVE AGENT	Drafts disclosures against each framework's specific requirements while preserving source attribution and prompt version metadata.
CONTROL & LOGGING LAYER	Immutable logs, service-account governance, human-in-the-loop workflows, and an exportable assurance pack ready to hand to an auditor.

For multinationals reporting into more than one regime, it is worth adding a regulatory routing layer on top of this stack — one that tailors disclosures per jurisdiction from a single, centrally controlled data model, rather than maintaining parallel spreadsheets for each framework.

REGIONAL CONTEXT

What this means for Pakistan and other emerging markets

For Pakistani exporters, conglomerates, and financial institutions connected to EU or multinational buyers, four points deserve particular attention. First, the value chain cap: a company with 1,000 employees or fewer cannot lawfully be required, as part of a customer's CSRD reporting process, to disclose more than the VSME standard covers — though buyers can still request more as a commercial matter, separate from CSRD compliance. Second, data quality now matters more than data volume: the practical bar

for a supplier is a small set of credible, auditable metrics — metering data, utility bills, logistics records, a supplier survey — rather than a lengthy narrative report. Third, AI bias and data gaps are a real constraint: poor-quality, unstandardised, self-reported data can undermine any algorithmic assessment built on top of it, making documented human-in-the-loop review non-negotiable. Fourth, investor scrutiny does not stop at national borders: as more asset managers apply AI to compare disclosures across companies and years, inconsistency against peers becomes a financing risk, not just a reputational one.

For firms already positioned around ISAE 3000 and ISAE 3410 assurance work, this is also where the commercial opportunity sits: companies that need a defensible, assurance-ready sustainability disclosure need exactly the combination of AI-literate data pipelines and independent assurance expertise described in this briefing.

GETTING STARTED

A 90-day roadmap to AI-ready ESG disclosure

Days 1–30 Inventory and lineage Map every reported metric back to its source system or document. Stand up immutable logging for all ESG data pipelines before any further automation is added.	Days 31–60 Controls and human review Implement versioning for prompts, models, and emission-factor libraries. Define the variance thresholds that trigger mandatory human sign-off on high-variance data.	Days 61–90 Continuous validation and assurance pack Schedule ongoing tests for accuracy, drift, and completeness rather than a year-end review. Produce one exportable audit trail from source to narrative.
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BOTTOM LINE

Governed automation, not speed, is the point

AI in ESG reporting stopped being primarily about drafting speed once regulators, auditors, and investors all started treating sustainability data the way they treat financial data. The companies that build in lineage, versioning, human-in-the-loop review, and continuous validation from the start will meet CSRD and ISSB expectations more easily and improve their standing with investors who are, right now, applying AI of their own to read what those companies publish. The companies that treat AI as a black box that simply produces polished paragraphs are the ones most likely to face assurance findings — and investor scepticism — sooner rather than later.

"Every reported metric must be reconstructable: source data → transformation → factor → formula → final value." That single sentence is, in effect, the whole test an auditor will apply to an

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SELECTED SOURCES

Further reading

1. Crowell & Moring, "EU Sustainability Reporting Revamp: Key Updates to the CSRD and the CS3D from the Omnibus I Directive" — crowell.com
2. CSRD Tools, "Omnibus: CSRD scope, thresholds and assurance after Directive (EU) 2026/470" — csrd-tools.com
3. Deloitte DART, "European Sustainability Reporting — Omnibus Legislative Developments" — dart.deloitte.com
4. BCLP, "CSRD: Requesting Data from SME Suppliers" — bclplaw.com
5. European Commission, "Feedback on sustainability reporting standards: additional explanatory information regarding the value chain cap" — finance.ec.europa.eu
6. S&P Global Sustainable1, "Where does the world stand on ISSB adoption?" — spglobal.com
7. Watershed, "A guide to California's climate disclosure rules (SB 253, SB 261, SB 219)" — watershed.com
8. Persefoni, "California SB 253 and SB 261: What Businesses Need to Know" — persefoni.com
9. ESG News, "New Clarity AI Survey Reveals AI's Growing Role as Climate Risk and Regulation Reshape Markets" — esgnews.com
10. World Journal of Advanced Research and Reviews, "Artificial Intelligence Models for Detecting Greenwashing" — wjarr.com
11. arXiv, "Corporate Greenwashing Detection in Text — a Survey" — arxiv.org
12. Spectreco, "CSRD VSME Standard 2026: What SME Suppliers Need to Know" — spectreco.com

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